Which Sale Really Wins? Discounts, Tax, Delivery

Tracking #: 2026-11-13-NAT-MATH-PROMOS-46

Fictional store policies: 5% tax on discounted goods only; delivery not taxed

Policy	Jacket	Trousers	Promotion	Delivery
Store A	$80	$30	25% off jacket	$6
Store B	$72	$32	15% off jacket	FREE
B alternate	$72	$32	10% off ALL goods	FREE
B delivery ..	Same B prices	Same B trousers	15% off jacket	$4

1. Compute Store A: jacket discount, jacket sale price, goods subtotal, sales tax and final total. Write a labelled five-line receipt and explain why delivery comes last.

2. Compute Store B with the 15%-off-jacket promotion. Show the five-line receipt and prove exactly how much lower it is than Store A.

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3. A buyer’s budget is $100. State which store fits, by how much, and which is over, by how much. Write a two-sentence recommendation using final costs, not ad slogans.

4. Store B also offers 10% OFF ALL goods instead. Calculate the goods subtotal, tax and final cost for THAT promotion. Compare both B offers to the nearest cent and name the better one.

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5. Now Store B charges $4 for delivery, without changing the 15% jacket discount. Recalculate the final cost and identify whether the winning store changes. Show the exact difference.

6. Audit this false receipt: ’Store A: $110 - 25% of $110 + 5% tax + $6 delivery.’ Mark TWO policy mistakes, provide corrected equations, and write a general rule for comparing unfamiliar store promotions.